

Elasticity

Lec-7 Chapter 4

Price Elasticity of Demand

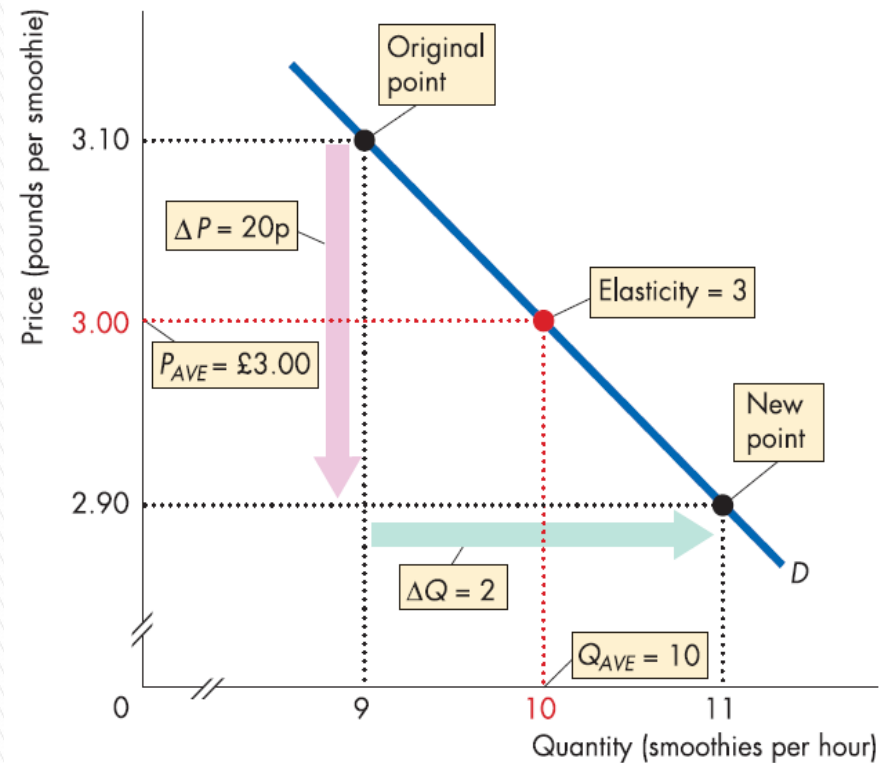
- ▶ The **price elasticity of demand** is a units-free measure of the responsiveness of the quantity demanded of a good or service, to a change in its price, when all other influences on buyers' plans remain the same.
- ▶ There can be differing degrees of elasticity or responsiveness of the quantity demanded to a change in price.

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

Price Elasticity of Demand

- ▶ $\Delta P/P_{AVE} = £(0.20/3.00) \times 100 = 6.67\%$
- ▶ $\Delta Q/Q_{AVE} = (2/10) \times 100 = 20\%$
- ▶ Price elasticity of demand = $20\% \div 6.67\% = 3$
- ▶ We use the average price and average quantity because it gives the most precise measurement of elasticity.

Calculating the Elasticity of Demand

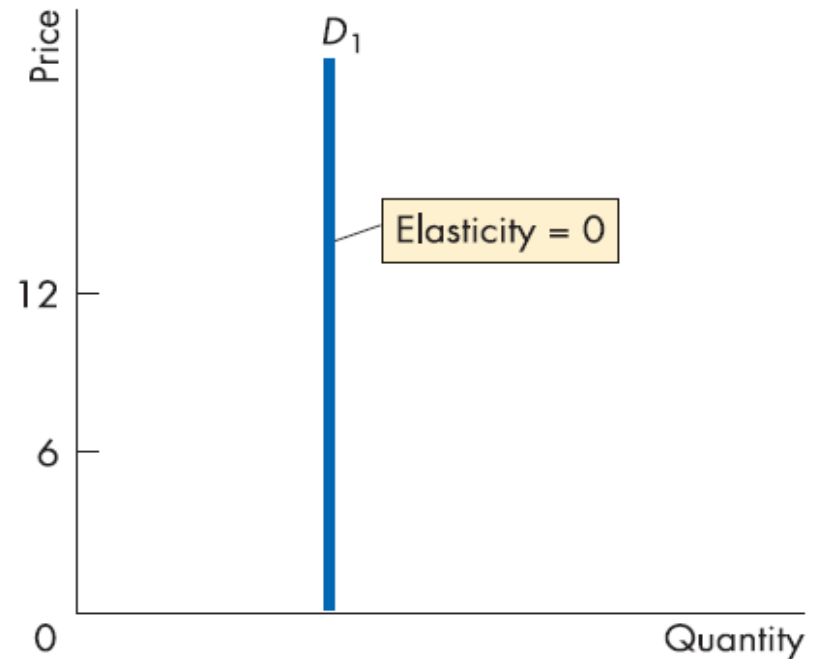


Price Elasticity of Demand

- ▶ When the price of a good rises, the quantity demanded decreases along the demand curve.
- ▶ A positive change in price brings a negative change in the quantity demanded, so the price elasticity of demand is a negative number.
- ▶ The magnitude, or absolute value, of the price elasticity of demand tells us how responsive or how elastic the demand is. We ignore the minus sign
- ▶ **Demand is inelastic if $PED < 1$**
- ▶ **Demand is elastic if $PED > 1$**

Inelastic and Elastic Demand: Range of possible elasticities of demand

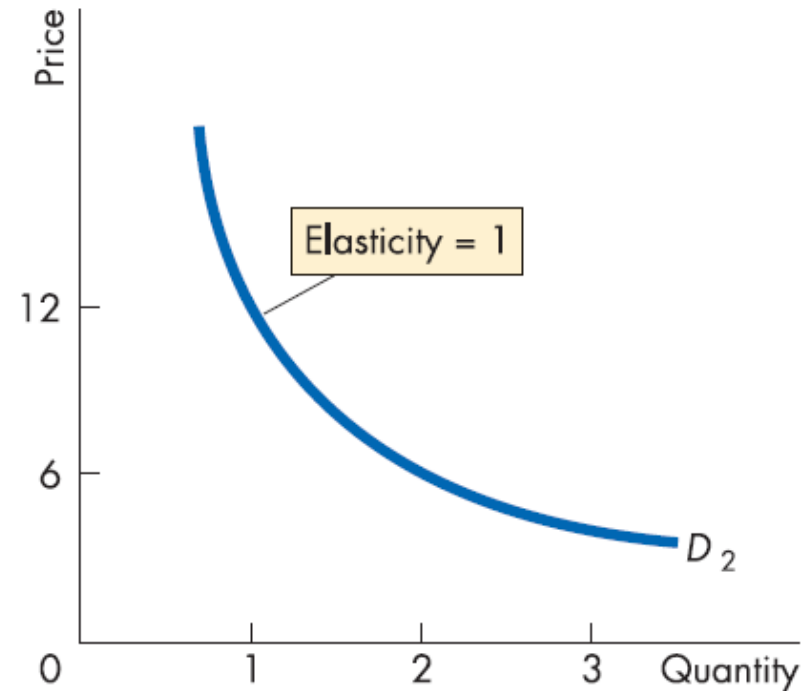
- ▶ **Perfectly inelastic demand**– If the quantity demanded remains constant when the price changes, then the price elasticity of demand is zero and the good is said to have a perfectly inelastic demand.



(a) Perfectly inelastic demand

Inelastic and Elastic Demand: Range of possible elasticities of demand

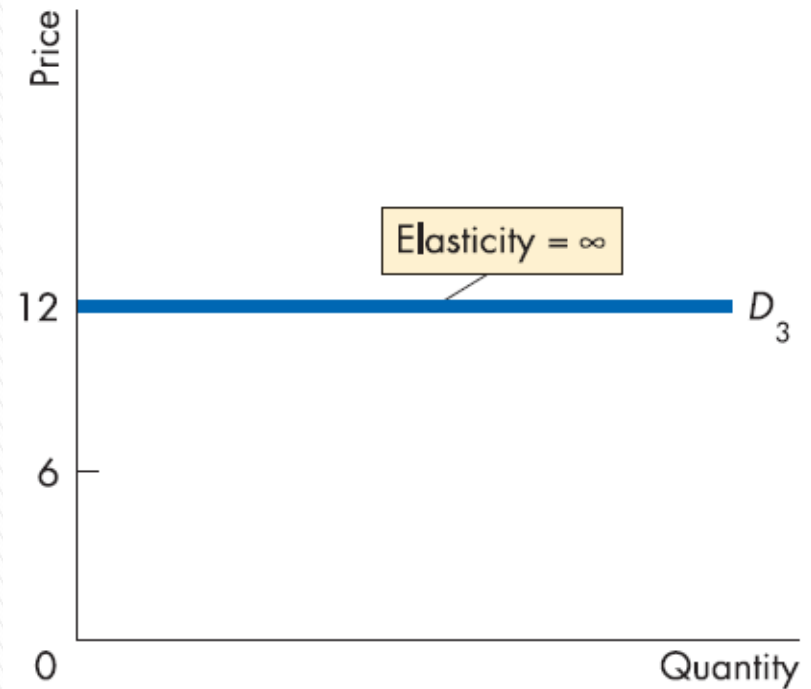
- ▶ **Unit Elastic Demand**– If the percentage change in the quantity demanded equals the percentage change in price, then the price elasticity equals 1 and the good is said to have a **unit elastic demand**



(b) Unit elastic demand

Inelastic and Elastic Demand: Range of possible elasticities of demand

- ▶ **Perfectly elastic demand**– If the quantity demanded changes by an infinitely large percentage in response to a tiny price change, then the price elasticity of demand is infinity and the good is said to have a **perfectly elastic demand**

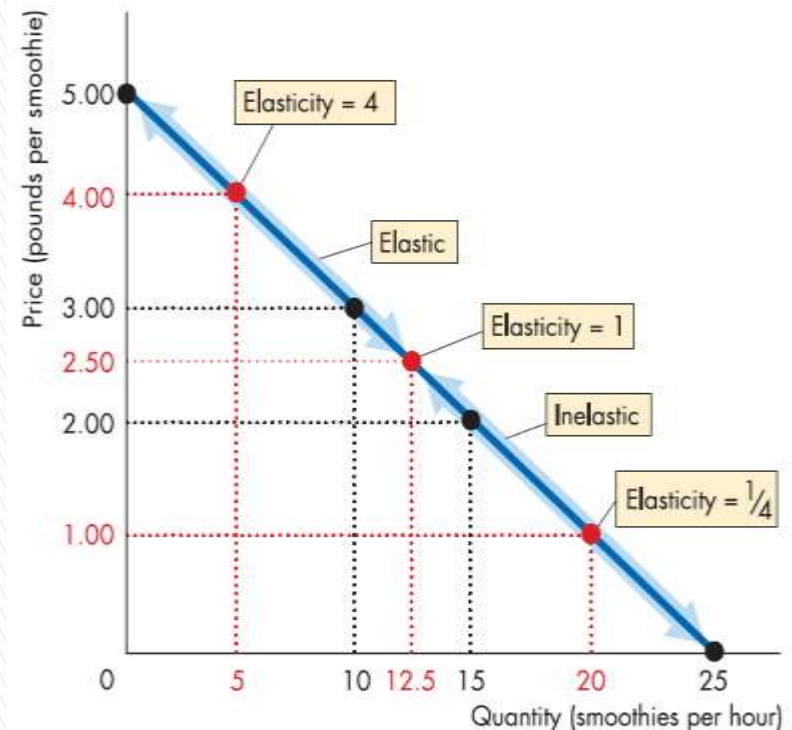


(c) **Perfectly elastic demand**

Inelastic and Elastic Demand

- ▶ Elasticity changes along a straight-line demand curve.
- ▶ At the mid-point of the curve, demand is unit elastic.
- ▶ At prices above the midpoint, demand is elastic.
- ▶ At prices below the midpoint, demand is inelastic.

Elasticity Along a Linear Demand Curve



Elasticity Along a Linear Demand Curve

Total Revenue and Elasticity

- ▶ The **total revenue from the sale of a good equals the** price of the good multiplied by the quantity sold.
- ▶ When a price changes, total revenue might or might not change.
- ▶ TR depends on the PED in the following ways:
 1. **If demand is elastic, a 1% price cut increases the quantity sold by more than 1% and total revenue increases.**
 2. **If demand is inelastic, a 1% price cut increases the quantity sold by less than 1% and total revenue decreases.**
 3. **If demand is unit elastic, a 1% price cut increases the quantity sold by 1% and so total revenue does not change.**

Total Revenue and Elasticity

- ▶ The **total revenue test** is a method of estimating the price elasticity of demand by observing the change in total revenue that results from a change in the price, when all other influences on the quantity sold remain the same.
 - If a price cut increases total revenue, demand is elastic.
 - If a price cut decreases total revenue, demand is inelastic.
 - If a price cut leaves total revenue unchanged, demand is unit elastic.
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The Factors That Influence the Elasticity of Demand

- ▶ The magnitude of the elasticity of demand depends on:
 1. The closeness of substitutes
 2. The proportion of income spent on the good
 3. The time elapsed since a price change
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The Factors That Influence the Elasticity of Demand

- ▶ **The closeness of substitutes**—The closer the substitutes for a good or service, the more elastic is the demand for it.
 - ▶ The degree of substitutability between two goods also depends on how narrowly (or broadly) we define them.
 - ▶ The elasticity of demand for personal computers is low, but the elasticity of demand for a Gateway or Dell PC is high.
 - ▶ Necessities
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The Factors That Influence the Elasticity of Demand

- ▶ **The proportion of income spent on the good–**
- ▶ Other things remaining the same, the greater the proportion of income spent on a good, the more elastic is the demand for it.
- ▶ Example– demand for toothpaste and housing.

The Factors That Influence the Elasticity of Demand

- ▶ **Time Elapsed Since Price Change**
 - ▶ The longer the time that has elapsed since a price change, the more elastic is demand.
 - ▶ Example– When the price of oil increased by 400 per cent during the 1970s, people barely changed the quantity of oil and petrol they bought. But gradually, as more efficient auto and airplane engines were developed, the quantity used decreased.
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Cross Elasticity of Demand

- ▶ The **cross elasticity of demand** is a measure of the responsiveness of the demand for a good to a change in the price of a substitute or complement, other things remaining the same.
- ▶ The cross elasticity of demand is positive for a substitute and negative for a complement

$$\text{Cross elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price of substitute or complement}}$$

Substitutes

- ▶ Suppose price of coffee is £1.5.
- ▶ 9 smoothies are sold per hour.
- ▶ Price of coffee rises to £2.5, there is no change in the price of smoothies.
- ▶ Quantity of smoothies sold increases to 11 an hour.
- ▶ $\Delta Q/Q_{AVG} = (2/10) \times 100 = 20\%$
- ▶ $\Delta P/P_{AVG} = £(1/2) \times 100 = 50\%$
- ▶ $XED = +20\% \div +50\% = 0.4$

Complements

- ▶ Now suppose that the price of smoothie is constant and 9 smoothies are sold an hour.
- ▶ Then the price of a salad falls from £2.50 to £1.50. No other influence on buying plans changes and the quantity of smoothies sold increases to 11 an hour.
- ▶ Smoothies and salads are complements, when the price of a salad falls, the demand for smoothies increases.
- ▶ Cross Elasticity of demand = $-20\% \div +50\% = -0.4$

Substitutes and Complements

- ▶ The magnitude of the cross elasticity of demand determines how far the demand curve shifts.
 - ▶ The larger the cross elasticity (absolute value), the greater is the change in demand and the larger is the shift in the demand curve.
 - ▶ If two items are very close substitutes or two items are close complements, the cross elasticity is large.
 - ▶ Two unrelated items have cross elasticity of demand equals to zero.
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Income Elasticity of Demand

- ▶ **Income Elasticity of Demand** measures the responsiveness of the demand for a good or service to a change in income, other things remaining the same.

$$\text{Income elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}$$

Income Elasticity of Demand

- ▶ Income elasticities of demand can be positive or negative and fall into three ranges:
 1. Greater than 1 (normal good, income elastic)
 2. Positive and less than 1 (normal good, income inelastic)
 3. Negative (inferior good)
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Income Elasticity of Demand

- ▶ Suppose that the price of smoothie is constant and 9 smoothies an hour are sold.
- ▶ Then income rise from £475 to £525 a week. No other influence on buying plans changes and the quantity of smoothies sold increases to 11 an hour.
- ▶ $20\% \div 10\% = 2$
- ▶ As income increases, the quantity of smoothies demanded increases faster than income. The demand for smoothies is income elastic.

Income Inelastic Demand

- ▶ If the % increase in the quantity demanded is less than the % increase in income, the income elasticity of demand is positive and less than 1. In this case, the quantity demanded increases as income increases, but income increases faster than the quantity demanded.
- ▶ **Inferior Goods**– If the quantity demanded of a good decreases when income increases, the income elasticity of demand is negative.

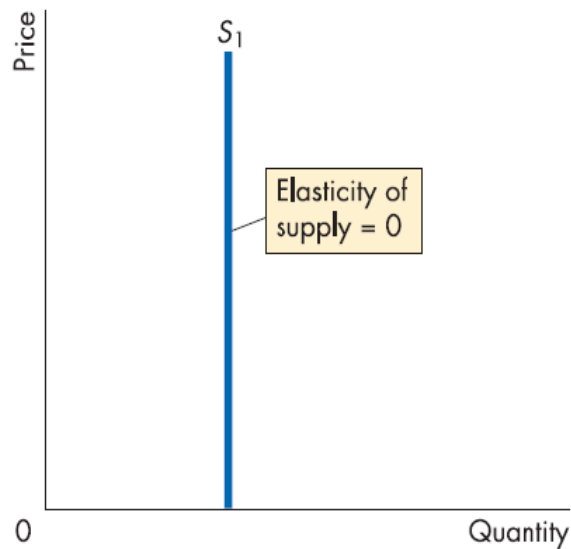
Elasticity of Supply

- ▶ The **elasticity of supply** measures the responsiveness of the quantity supplied to a change in the price of a good when all other influences on selling plans remain the same.

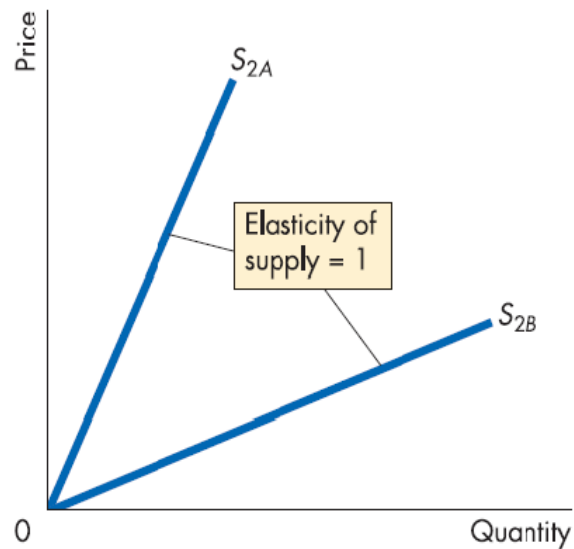
$$\text{Price elasticity of supply} = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

Range of Elasticities of Supply

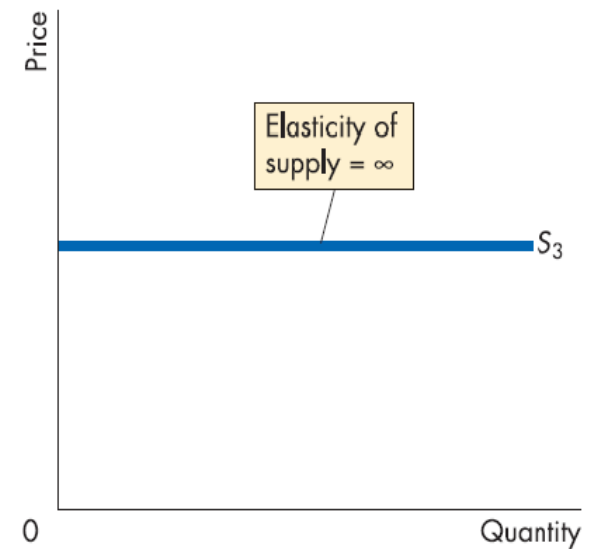
Inelastic and Elastic Supply



(a) Perfectly inelastic supply



(b) Unit elastic supply



(c) Perfectly elastic supply

Factors Influencing the Elasticity of Supply

- ▶ The magnitude of the elasticity of supply depends on:
 1. Resource substitution possibilities
 2. Time frame for the supply decision
- ▶ **Resource Substitution Possibilities:** goods and services that require unique productive resources have a low or a zero elasticity of supply.

Goods and services that can be produced using commonly available resources that could be allocated to a wide variety of alternative tasks, have a high elasticity of supply

Factors Influencing the Elasticity of Supply

▶ Time frame for the supply decision

1. Momentary supply
 2. Long-run supply
 3. Short-run supply
- ▶ The **momentary supply curve** shows the response of the quantity supplied immediately following a price change.
 - ▶ The **long-run supply** curve shows the response of the quantity supplied to a change in price after all the technologically possible ways of adjusting supply have been exploited.
 - ▶ The **short-run supply** curve shows how the quantity supplied responds to a price change when only some of the technologically possible adjustments to production have been made.

Parkin, Powell, Matthews.
Economics. 8th Edition
Chapter 4

